Series

$$\oplus\ e^{u}=\sum_{n=0}^{\infty}\left(\frac{x^{n}}{n!}\right)=1+x+\frac{x^{2}}{2!}+\frac{x^{3}}{3!}+\frac{x^{4}}{4!}+\dots$$

$$\oplus\ \log(x)=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}x^{n}}{n}\right)=x-\frac{x^{2}}{2}+\frac{x^{3}}{3}-\frac{x^{4}}{4}+\dots$$

$$\oplus\,(x+y)^{z}=\sum_{i=0}^{z}\binom{z}{i}x^{z-i}y^{i}=\binom{z}{0}x^{z}+\binom{z}{1}x^{z-1}y+\dots+\binom{z}{z-1}x.y^{z-1}+\binom{z}{z}y^{z}$$

$$\oplus\ \frac{1}{1-x}=\sum_{n=0}^{\infty}\left(x^{n}\right)=1+x+x^{2}+x^{3}+\dots\qquad ,|x|<1$$

$$\oplus\ \frac{1}{(1-x)^{n}}=\sum_{k=0}^{\infty}x^{k}C(n+k-1,k)=1+xC(n,1)+x^{2}C(n+1,2)+\dots\qquad ,|x|<1$$

$$\oplus\ \log(1-x)=-\sum_{n=1}^{\infty}\left(\frac{x^{n}}{n}\right)=-x-\frac{x^{2}}{2}-\frac{x^{3}}{3}-\dots$$

$$\oplus\ \log(1+x)=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n+1}x^{n}}{n}\right)=x-\frac{x^{2}}{2}+\frac{x^{3}}{3}-\frac{x^{4}}{4}+\dots$$

$$\oplus\ \log\left(\frac{1+x}{1-x}\right)=2\sum_{n=0}^{\infty}\left(\frac{x^{2n+1}}{2n+1}\right)=2\left(x+\frac{x^{3}}{3}+\frac{x^{5}}{5}+\frac{x^{7}}{7}+\dots\right)$$

$$\oplus\ \cos(x)=\sum_{n=0}^{\infty}\left(\frac{(-1)^{n}x^{2n}}{(2n)!}\right)=1-\frac{x^{2}}{2!}+\frac{x^{4}}{4!}-\frac{x^{6}}{6!}+\dots$$

$$\oplus\ \sin(x)=\sum_{n=0}^{\infty}\left(\frac{(-1)^{n}x^{2n+1}}{(2n+1)!}\right)=x-\frac{x^{3}}{3!}+\frac{x^{5}}{5!}-\frac{x^{7}}{7!}+\dots$$

$$\oplus\ \tan(x)=\sum_{n=1}^{\infty}\left(\frac{B_{2n}(-4)^{n}(1-4^{n})x^{2n-1}}{(2n)!}\right)=x+\frac{x^{3}}{3}+\frac{2x^{5}}{15}+\dots\quad ,|x|<\frac{\pi}{2}$$

$$\oplus\ \sec(x)=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n}x^{2n}E_{2n}}{(2n)!}\right)=1+\frac{x^{2}}{2}+\frac{5x^{4}}{24}+\dots\quad ,|x|<\frac{\pi}{2}$$

$$\oplus\ \sin^{-1}(x)=\sum_{n=0}^{\infty}\left(\frac{(2n)!x^{2n+1}}{(2n+1)(n!)^{2}4^{n}}\right)=x+\frac{x^{3}}{6}+\frac{3x^{5}}{40}+\dots\qquad ,|x|\le 1$$

$$\oplus\ \cos^{-1}(x)=\frac{\pi}{2}-\sum_{n=0}^{\infty}\left(\frac{(2n)!x^{2n+1}}{(2n+1)(n!)^{2}4^{n}}\right)=\frac{\pi}{2}-x-\frac{x^{3}}{6}-\frac{3x^{5}}{40}-\dots\qquad ,|x|\le 1$$

$$\oplus\ \tan^{-1}(x)=\sum_{n=0}^{\infty}\left(\frac{(-1)^{n}x^{2n+1}}{2n+1}\right)=x-\frac{x^{3}}{3}+\frac{x^{5}}{5}-\frac{x^{7}}{7}+\dots\quad ,,|x|\le 1$$

$$\oplus\ \cosh(x)=\sum_{n=0}^{\infty}\left(\frac{x^{2n}}{(2n)!}\right)=1+\frac{x^2}{2!}+\frac{x^4}{4!}+\frac{x^6}{6!}+\ldots$$

$$\oplus\ \sinh(x)=\sum_{n=0}^{\infty}\left(\frac{x^{2n+1}}{(2n+1)!}\right)=x+\frac{x^3}{3!}+\frac{x^5}{5!}+\frac{x^7}{7!}+\ldots$$

$$\oplus\ \tanh(x)=\sum_{n=0}^{\infty}\left(\frac{B_{2n}4^n(4^n-1)x^{2n-1}}{(2n)!}\right)=x-\frac{x^3}{3}+\frac{2x^5}{15}-\frac{17x^7}{315}+\ldots\qquad ,|x|<\frac{\pi}{2}$$

$$\oplus\ \sinh^{-1}(x)=\sum_{n=0}^{\infty}\left(\frac{(-1)^n(2n)!x^{2n+1}}{4^n(n!)^2(2n+1)}\right)=x-\frac{x^3}{6}+\frac{3x^5}{40}-\ldots\qquad ,\ |x|\le 1$$

$$\oplus\ \tanh^{-1}(x)=\sum_{n=0}^{\infty}\left(\frac{x^{2n+1}}{(2n+1)}\right)=x+\frac{x^3}{3}+\frac{x^5}{5}+\ldots\qquad ,\left(|x|\le 1\right)$$

$$\oplus\ H_n=\sum_{k=1}^{n}\left(\frac{1}{k}\right)=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\ldots\frac{1}{n}$$

$$\oplus B_m=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^m}\right)=1-\frac{1}{3^m}+\frac{1}{5^m}-\frac{1}{7^m}+\ldots$$

$$\oplus\ Li_2(x)=\sum_{n=1}^{\infty}\left(\frac{x^n}{n^2}\right)=x+\frac{x^2}{2^2}+\frac{x^3}{3^2}+\frac{x^4}{4^2}+\ldots$$

$$\oplus\ Li_3(x)=\sum_{n=1}^{\infty}\left(\frac{x^n}{n^3}\right)=x+\frac{x^2}{2^3}+\frac{x^3}{3^3}+\frac{x^4}{4^3}+\ldots$$

$$\oplus\ G=\sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n+1)^2}\right)=1-\frac{1}{3^2}+\frac{1}{5^2}-\frac{1}{7^2}+\ldots\qquad \text{``}G\text{'' Catalan's constant}$$

$$\oplus W(x)=\sum_{n=1}^{\infty}\left(\frac{(-n)^{n-1}}{n!}.x^n\right)=x-x^2+\frac{3x^3}{2}-\frac{16x^4}{6}+\ldots\qquad \text{``}W\text{'' Lambert function}$$

$$\oplus n!!=\begin{cases} n(n-2)\ldots(6)(4)(2) & ,If:\text{``}n\text{''}\ even\quad ;Ex:8!!=8\times6\times4\times2\\ n(n-2)\ldots(5)(3)(1) & ,If:\text{``}n\text{''}\ odd\quad ;Ex:7!!=7\times5\times3\times1\end{cases}$$

$$\oplus\ \eta(s)=\sum_{n=1}^{\infty}\left(\frac{(-1)^{n-1}}{n^s}\right)=1-\frac{1}{2^s}+\frac{1}{3^s}-\cdots\qquad \oplus\ \zeta(s)=\sum_{n=1}^{\infty}\left(\frac{1}{n^s}\right)=1+\frac{1}{2^s}+\frac{1}{3^s}+\cdots$$

$$\oplus\ \ 1+2+3+\cdots+n=\frac{n(n+1)}{2}$$

$$\oplus\ \ 1^2+2^2+3^2+\cdots+n^2=\frac{n(n+1)(2n+1)}{6}$$

$$\oplus\ \ 1^3+2^3+3^3+\cdots+n^3=\left[\frac{n(n+1)}{2}\right]^2$$

EXERCICES ET SOLUTION

001. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1+2+3+\cdots+n}{n^2}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1+2+3+\cdots+n}{n^2}\right)$

$$=\frac{1}{2}\lim_{n\to+\infty}\left(\frac{n(n+1)}{n^2}\right)$$

$$1+2+3+\cdots+n=\frac{n(n+1)}{2}$$

$$=\frac{1}{2}\lim_{n\to+\infty}\left(1+\frac{1}{n}\right)=\frac{1}{2}$$

ដូចនេះ $L=\dfrac{1}{2}$ ។

002. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1^2+2^2+3^2+\cdots+n^2}{n^3}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1^2+2^2+3^2+\cdots+n^2}{n^3}\right)$

$$1^3+2^3+3^3\cdots+n^3=\frac{n(n+1)(2n+1)}{6}$$

$$=\frac{1}{6}\lim_{n\to+\infty}\left(\frac{n(n+1)(2n+1)}{n^3}\right)$$

$$=\frac{1}{6}\lim_{n\to+\infty}\left[\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right]$$

$$=\frac{1}{6}\times2=\frac{1}{3}$$ b

ដូចនេះ $L=\dfrac{1}{3}$ ។

003. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1}{2}+\dfrac{1}{4}+\dfrac{1}{8}+\cdots+\dfrac{1}{2^n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1}{2}+\dfrac{1}{4}+\dfrac{1}{8}+\cdots+\dfrac{1}{2^n}\right)$

$$=\lim_{n\to+\infty}\left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots+\frac{1}{2^n}\right)$$

$$=\lim_{n\to+\infty}\left(\frac{1}{2}\times\frac{1-\frac{1}{2^n}}{1-\frac{1}{2}}\right)=\frac{1}{2}\times\frac{1-0}{\frac{1}{2}}=1$$

ដូចនេះ $L=1$ ។

004. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\frac{1}{k(k+1)}\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\frac{1}{k(k+1)}\right)\right]$

នោះយើងបាន $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\frac{1}{k}-\frac{1}{k+1}\right)\right]$

$$=\lim_{n\to+\infty}\left(1-\frac{1}{n+1}\right)=1$$

ដូចនេះ $L=1$ ។

005. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\prod\limits_{k=2}^{n}\left(\frac{k^3-1}{k^3+1}\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\prod\limits_{k=2}^{n}\left(\frac{k^3-1}{k^3+1}\right)\right]$

យើងមានទំនាក់ទំនង $\left(\frac{k^3-1}{k^3+1}\right)=\left(\frac{k-1}{k+1}\right)\left(\frac{k^2+k+1}{k^2-k+1}\right)$

យើងបាន $L=\lim\limits_{n\to+\infty}\left[\prod\limits_{k=2}^{n}\left(\frac{k-1}{k+1}\right)\left(\frac{k^2+k+1}{k^2-k+1}\right)\right]$

$$=\lim_{n\to+\infty}\left[\prod_{k=2}^{n}\left(\frac{k-1}{k+1}\right)\prod_{k=2}^{n}\left(\frac{k^2+k+1}{k^2-k+1}\right)\right]$$

$$=\lim_{n\to+\infty}\left(\frac{1\cdot 2}{n(n+1)}\times\frac{n^2+n+1}{3}\right)=\frac{2}{3}$$

ដូចនេះ $L=\frac{2}{3}$ ។

006. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=3}^{n}\left(\dfrac{k}{(k-2)!+(k-1)!+k!}\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=3}^{n}\left(\dfrac{k}{(k-2)!+(k-1)!+k!}\right)\right]$

ដោយ $\dfrac{k}{(k-2)!+(k-1)!+k!}=\dfrac{k}{(k-2)!\left[1+(k-1)+(k-1)k\right]}=\dfrac{1}{(k-2)!\cdot k}$

$$=\frac{k-1}{(k-2)!\cdot(k-1)\cdot k}=\frac{k-1}{k!}$$

$$=\frac{k}{k!}-\frac{1}{k!}=\frac{1}{(k-1)!}-\frac{1}{k!}$$

យើងបាន $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=3}^{n}\left(\dfrac{1}{(k-1)!}-\dfrac{1}{k!}\right)\right]$

$$=\lim_{n\to+\infty}\left(\frac{1}{2!}-\frac{1}{n!}\right)=\frac{1}{2}$$

ដូចនេះ $L=\dfrac{1}{2}$ ។

007. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\dfrac{k}{k^4+k^2+1}\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\dfrac{k}{k^4+k^2+1}\right)\right]$

ដោយ $\dfrac{k}{k^4+k^2+1}=\dfrac{k}{(k^2+k+1)(k^2-k+1)}$

$$=\frac{1}{2}\cdot\frac{(k^2+k+1)-(k^2-k+1)}{(k^2+k+1)(k^2-k+1)}$$

$$=\frac{1}{2}\left(\frac{1}{k^2-k+1}-\frac{1}{k^2+k+1}\right)$$

យើងបាន $L=\lim\limits_{n\to+\infty}\left[\dfrac{1}{2}\sum\limits_{k=1}^{n}\left(\dfrac{1}{k^2-k+1}-\dfrac{1}{k^2+k+1}\right)\right]$

$$=\frac{1}{2}\lim_{n\to+\infty}\left(1-\frac{1}{n^2+n+1}\right)=\frac{1}{2}$$

ដូចនេះ $L=\dfrac{1}{2}$ ។

008. ចូរគណនាលីមីត $L=\lim\limits_{x\to1}\left(\dfrac{x+x^2+x^3+\cdots+x^n-n}{x-1}\right)\quad,\forall n\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត
$$
\begin{aligned}
L&=\lim_{x\to1}\left(\frac{x+x^2+x^3+\cdots+x^n-n}{x-1}\right)\quad,\forall n\in\mathbb{N}\\
&=\lim_{x\to1}\left[\frac{(x-1)+(x^2-1)+(x^3-1)+\cdots+(x^n-1)}{(x-1)}\right]\\
&=\lim_{x\to1}\left[\frac{(x-1)\left[1+(x+1)+(x^2+x+1)+\cdots+(x^{n-1}+x^{n-2}+\cdots+x+1)\right]}{(x-1)}\right]\\
&=\lim_{x\to1}\left[1+(x+1)+(x^2+x+1)+\cdots+(x^{n-1}+x^{n-2}+\cdots+x+1)\right]\\
&=\left(1+2+3+\cdots+n\right)=\frac{n(n+1)}{2}
\end{aligned}
$$

ដូចនេះ $L=\dfrac{n(n+1)}{2}$ ។

009. ចូរគណនាលីមីត $L=\lim\limits_{x\to1}\left(\dfrac{x+x^2+x^3+\cdots+x^p-p}{x+x^2+x^3+\cdots+x^q-q}\right)\quad,\forall p\,\&\,q\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត
$$
\begin{aligned}
L&=\lim_{x\to1}\left(\frac{x+x^2+x^3+\cdots+x^p-p}{x+x^2+x^3+\cdots+x^q-q}\right)\quad,\forall p\,\&\,q\in\mathbb{N}\\
&=\lim_{x\to1}\left(\frac{(x-1)+(x^2-1)+(x^3-1)+\cdots+(x^p-1)}{(x-1)+(x^2-1)+(x^3-1)+\cdots+(x^q-1)}\right)\\
&=\lim_{x\to1}\left[\frac{(x-1)\left[1+(x+1)+(x^2+x+1)+\cdots+(x^{p-1}+x^{p-2}+\cdots+x+1)\right]}{(x-1)\left[1+(x+1)+(x^2+x+1)+\cdots+(x^{q-1}+x^{q-2}+\cdots+x+1)\right]}\right]\\
&=\lim_{x\to1}\left[\frac{1+(x+1)+(x^2+x+1)+\cdots+(x^{p-1}+x^{p-2}+\cdots+x+1)}{1+(x+1)+(x^2+x+1)+\cdots+(x^{q-1}+x^{q-2}+\cdots+x+1)}\right]\\
&=\left(\frac{1+2+3+\cdots+p}{1+2+3+\cdots+q}\right)=\frac{p(p+1)}{q(q+1)}
\end{aligned}
$$

ដូចនេះ $L=\dfrac{p(p+1)}{q(q+1)}$ ។

010. ចូរគណនាលីមីត $L=\lim\limits_{x\to 0}\left(\dfrac{n-\cos(x)-\cos^2(x)-\cos^3(x)-\cdots-\cos^n(x)}{x^2}\right)\ ,\forall n\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to 0}\left(\dfrac{n-\cos(x)-\cos^2(x)-\cos^3(x)-\cdots-\cos^n(x)}{x^2}\right)\ ,\forall n\in\mathbb{N}$

$$=\lim_{x\to 0}\left[\frac{(1-\cos(x))\left[1+(1+\cos(x))+\cdots+\left(1+\cos(x)+\cdots+\cos^{n-1}(x)\right)\right]}{x^2}\right]$$

$$=\lim_{x\to 0}\left[\frac{1-\cos(x)}{x^2}\times\left[1+(1+\cos(x))+\cdots+\left(1+\cos(x)+\cdots+\cos^{n-1}(x)\right)\right]\right]$$

$$=\frac{1}{2}\times(1+2+3+\cdots+n)=\frac{n(n+1)}{4}$$

ដូចនេះ: $L=\dfrac{n(n+1)}{4}$ ។

011. ចូរគណនាលីមីត $L=\lim\limits_{x\to 0}\left(\dfrac{n-\cos(x)-\cos(2x)-\cos(3x)-\cdots-\cos(nx)}{x^2}\right)\ ,\forall n\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to 0}\left(\dfrac{n-\cos(x)-\cos(2x)-\cos(3x)-\cdots-\cos(nx)}{x^2}\right)$

$$=\lim_{x\to 0}\left(\frac{1-\cos(x)+1-\cos(2x)+1-\cos(3x)+\cdots+1-\cos(nx)}{x^2}\right)$$

$$=\lim_{x\to 0}\left(\frac{1-\cos(x)}{x^2}+\frac{1-\cos(2x)}{x^2}+\frac{1-\cos(3x)}{x^2}+\cdots+\frac{1-\cos(nx)}{x^2}\right)$$

$$=\frac{1}{2}+\frac{2^2}{2}+\frac{3^2}{2}+\cdots+\frac{n^2}{2}=\frac{n(n+1)(2n+1)}{12}$$

ដូចនេះ: $L=\dfrac{n(n+1)(2n+1)}{12}$ ។

0012. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1}{\Gamma(3)}+\dfrac{2}{\Gamma(4)}+\dfrac{3}{\Gamma(5)}+\cdots+\dfrac{n}{\Gamma(n+2)}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1}{\Gamma(3)}+\dfrac{2}{\Gamma(4)}+\dfrac{3}{\Gamma(5)}+\cdots+\dfrac{n}{\Gamma(n+2)}\right)$

$$=\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{k}{\Gamma(k+2)}\right)\right]=\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{k}{(k+1)!}\right)\right]$$

យើងមានទំនាក់ទំនង $\dfrac{k}{(k+1)!}=\dfrac{k+1-1}{(k+1)!}$

$$=\frac{k+1}{(k+1)!}-\frac{1}{(k+1)!}$$

$$=\frac{1}{k!}-\frac{1}{(k+1)!}$$

នោះយើងបាន $L=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\dfrac{1}{k!}-\dfrac{1}{(k+1)!}\right)\right]$

$$=\lim_{n\to+\infty}\left(\frac{1}{1!}-\frac{1}{(n+1)!}\right)=1$$

ដូចនេះ $L=1$ ។

013. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1}{1\cdot2\cdot3}+\dfrac{1}{2\cdot3\cdot4}+\cdots+\dfrac{1}{n\cdot(n+1)\cdot(n+2)}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1}{1\cdot2\cdot3}+\dfrac{1}{2\cdot3\cdot4}+\cdots+\dfrac{1}{n\cdot(n+1)\cdot(n+2)}\right)$

$$=\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{1}{k\cdot(k+1)\cdot(k+2)}\right)\right]$$

ដោយ $\dfrac{1}{k\cdot(k+1)\cdot(k+2)}=\dfrac{1}{2}\cdot\dfrac{(k+2)-k}{k\cdot(k+1)\cdot(k+2)}$

$$=\frac{1}{2}\left(\frac{1}{k\cdot(k+1)}-\frac{1}{(k+1)\cdot(k+2)}\right)$$

នោះយើងបាន $L=\dfrac{1}{2}\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\dfrac{1}{k\cdot(k+1)}-\dfrac{1}{(k+1)\cdot(k+2)}\right)\right]$

$$=\frac{1}{2}\lim_{n\to+\infty}\left(\frac{1}{1\cdot2}-\frac{1}{(n+1)(n+2)}\right)=\frac{1}{4}$$

ដូចនេះ $L=\dfrac{1}{4}$ ។

014. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{n}{\sqrt{n^4+1}}+\dfrac{n}{\sqrt{n^4+2}}+\dfrac{n}{\sqrt{n^4+3}}+\cdots+\dfrac{n}{\sqrt{n^4+n}}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{n}{\sqrt{n^4+1}}+\dfrac{n}{\sqrt{n^4+2}}+\dfrac{n}{\sqrt{n^4+3}}+\cdots+\dfrac{n}{\sqrt{n^4+n}}\right)$

$$=\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{n}{\sqrt{n^4+k}}\right)\right]$$

ចំពោះគ្រប់ $k\in\mathbb{N}\,,k\le n$ ដែល $n\in\mathbb{N}$

យើងបាន $n^4+1\le n^4+k\le n^4+n$

សមមូលឱ្យ $\dfrac{n}{\sqrt{n^4+1}}\le\dfrac{n}{\sqrt{n^4+k}}\le\dfrac{n}{\sqrt{n^4+n}}$

$$\sum_{k=1}^{n}\left(\frac{n}{\sqrt{n^4+1}}\right)\le\sum_{k=1}^{n}\left(\frac{n}{\sqrt{n^4+k}}\right)\le\sum_{k=1}^{n}\left(\frac{n}{\sqrt{n^4+n}}\right)$$

$$\lim_{n\to+\infty}\left(\frac{n^2}{\sqrt{n^4+1}}\right)\le\lim_{n\to+\infty}\sum_{k=1}^{n}\left(\frac{n}{\sqrt{n^4+k}}\right)\le\lim_{n\to+\infty}\left(\frac{n^2}{\sqrt{n^4+n}}\right)$$

យើងបាន $1\le L\le 1$ យើងទាញបាន $L=1$

ដូចនេះ $L=1$ ។

015. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\sqrt{n+\sqrt{(n-1)+\sqrt{(n-2)+\cdots+\sqrt{3+\sqrt{2+\sqrt{1}}}}}}-\sqrt{n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\sqrt{n+\sqrt{(n-1)+\sqrt{(n-2)+\cdots+\sqrt{3+\sqrt{2+\sqrt{1}}}}}}-\sqrt{n}\right)$

តាមឯកលក្ខណៈភាព $\sqrt{a}-\sqrt{b}=\dfrac{(\sqrt{a}-\sqrt{b})(\sqrt{a}+\sqrt{b})}{(\sqrt{a}+\sqrt{b})}=\dfrac{a-b}{\sqrt{a}+\sqrt{b}}$ ដែល $a,b>0$

នោះយើងបាន $L=\lim\limits_{n\to+\infty}\left(\dfrac{n+\sqrt{(n-1)+\sqrt{(n-2)+\cdots+\sqrt{3+\sqrt{2+\sqrt{1}}}}}-n}{\sqrt{n+\sqrt{(n-1)+\sqrt{(n-2)+\cdots+\sqrt{3+\sqrt{2+\sqrt{1}}}}}}+\sqrt{n}}\right)$

$$=\lim_{n\to+\infty}\left(\frac{\sqrt{(n-1)+\sqrt{(n-2)+\cdots+\sqrt{3+\sqrt{2+\sqrt{1}}}}}}{\sqrt{n+\sqrt{(n-1)+\cdots+\sqrt{3+\sqrt{2+\sqrt{1}}}}}+\sqrt{n}}\right)$$

$$=\lim_{n\to+\infty}\left(\frac{\sqrt{n-1}}{\sqrt{n}+\sqrt{n}}\right)=\lim_{n\to+\infty}\left(\frac{\sqrt{n}}{2\sqrt{n}}\right)=\frac{1}{2}$$

ដូចនេះ: $L=\frac{1}{2}$ ។

016. ចូរគណនាលីមីត $L=\lim_{n\to0}n^2\left(1+2+3+\cdots+\left\lfloor\frac{1}{|n|}\right\rfloor\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim_{n\to0}n^2\left(1+2+3+\cdots+\left\lfloor\frac{1}{|n|}\right\rfloor\right)$

តាង $y=\frac{1}{|n|}\Rightarrow n^2=\frac{1}{y^2}$ បើ $n\to0$ នោះ $y\to\infty$

យើងបាន $L=\lim_{y\to\infty}\frac{\left(1+2+3+\cdots+\lfloor y\rfloor\right)}{y^2}=\frac{1}{2}\lim_{y\to\infty}\frac{\lfloor y\rfloor\left(\lfloor y\rfloor+1\right)}{y^2}$

ដោយ $y-1\le\lfloor y\rfloor\le y$ និង $y\le\lfloor y\rfloor+1\le y+1$ ដែល y ជាចំនួនពិត

សមមូលឱ្យ $\frac{1}{2}\times\frac{y(y-1)}{y^2}\le\frac{1}{2}\times\frac{\lfloor y\rfloor\left(\lfloor y\rfloor+1\right)}{y^2}\le\frac{1}{2}\times\frac{y(y+1)}{y^2}$

យើងបាន $\frac{1}{2}\lim_{y\to\infty}\left(\frac{y(y-1)}{y^2}\right)\le L\le\frac{1}{2}\lim_{y\to\infty}\left(\frac{y(y+1)}{y^2}\right)$

សមមូលឱ ្យ $\frac{1}{2}\le L\le\frac{1}{2}$

យើងទាញបាន៖ $L=\frac{1}{2}$

ដូចនេះ: $L=\frac{1}{2}$ ។

017. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\left(\frac{1}{\sqrt{3}+1}+\frac{1}{\sqrt{5}+\sqrt{3}}+\cdots+\frac{1}{\sqrt{2n+1}+\sqrt{2n-1}}\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\left(\frac{1}{\sqrt{3}+1}+\frac{1}{\sqrt{5}+\sqrt{3}}+\cdots+\frac{1}{\sqrt{2n+1}+\sqrt{2n-1}}\right)\right]$

$$=\lim_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\sum_{k=1}^{n}\left(\frac{1}{\sqrt{2k+1}+\sqrt{2k-1}}\right)\right]$$

ដោយ $\frac{1}{\sqrt{2k+1}+\sqrt{2k-1}}=\frac{(\sqrt{2k+1}-\sqrt{2k-1})}{(\sqrt{2k+1}-\sqrt{2k-1})(\sqrt{2k+1}-\sqrt{2k-1})}$

$$=\frac{1}{2}\left(\sqrt{2k+1}-\sqrt{2k-1}\right)$$

នោះយើងបាន $L=\frac{1}{2}\lim\limits_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\sum_{k=1}^{n}\left(\sqrt{2k+1}-\sqrt{2k-1}\right)\right]$

$$=\frac{1}{2}\lim_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\left(\sqrt{2n+1}-1\right)\right]$$

$$=\frac{1}{2}\lim_{n\to+\infty}\left(\sqrt{2+\frac{1}{n}}-\sqrt{\frac{1}{n}}\right)$$

$$=\frac{1}{2}\left(\sqrt{2+0}-\sqrt{0}\right)=1$$

ដូចនេះ $L=1$ ។

018. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[n\prod_{p=1}^{n}\left(\sum_{q=1}^{p}\left(\frac{1}{q(q+1)}\right)\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[n\prod_{p=1}^{n}\left(\sum_{q=1}^{p}\left(\frac{1}{q(q+1)}\right)\right)\right]$

ដោយ $\frac{1}{q(p+1)}=\frac{(q+1)-q}{q(p+1)}=\frac{1}{q}-\frac{1}{q+1}$

យើងបាន៖ $L=\lim\limits_{n\to+\infty}\left[n\prod_{p=1}^{n}\left(\sum_{q=1}^{p}\left(\frac{1}{q}-\frac{1}{q+1}\right)\right)\right]$

$$= \lim_{n\to+\infty}\left[n\prod_{p=1}^{n}\left(\frac{1}{1}-\frac{1}{2}+\frac{1}{2}-\frac{1}{3}+\cdots+\frac{1}{p}-\frac{1}{p+1}\right)\right]$$

$$= \lim_{n\to+\infty}\left[n\prod_{p=1}^{n}\left(1-\frac{1}{p+1}\right)\right]$$

$$= \lim_{n\to+\infty}\left[n\prod_{p=1}^{n}\left(\frac{p}{p+1}\right)\right]$$

$$= \lim_{n\to+\infty}\left[n\left(\frac{1}{2}\times\frac{2}{3}\times\cdots\times\frac{n}{n+1}\right)\right]$$

$$= \lim_{n\to+\infty}\left[n\left(\frac{1}{n+1}\right)\right]=1$$

ដូចនេះ $L=1$ ។

019. ចូរគណនាលីមីត $L=\lim\limits_{x\to1}\left(\dfrac{\prod_{k=1}^{n}(x^k-1)}{(x-1)^n}\right)$ $,\forall n\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to1}\left(\dfrac{\prod_{k=1}^{n}(x^k-1)}{(x-1)^n}\right)$ $,\forall n\in\mathbb{N}$

ដោយ $\prod\limits_{k=1}^{n}(x^k-1)=(x-1)(x^2-1)(x^3-1)\cdots(x^n-1)$

$$=(x-1)^n\left[(x+1)(x^2+x+1)\cdots(x^{n-1}+x^{n-2}+\cdots+1)\right]$$

យើងបាន $L=\lim\limits_{x\to1}\left(\dfrac{(x-1)^n\left[(x+1)(x^2+x+1)\cdots(x^{n-1}+x^{n-2}+\cdots+1)\right]}{(x-1)^n}\right)$

$$=\lim_{x\to1}\left((x+1)(x^2+x+1)\cdots(x^{n-1}+x^{n-2}+\cdots+1)\right)$$

$$=2\times3\times4\times\cdots\times n=n!$$

ដូចនេះ $L=n!$ ។

020. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1+2}+\frac{1}{1+2+3}+\frac{1}{1+2+3+4}+\cdots+\frac{1}{1+2+3+\cdots+n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1+2}+\frac{1}{1+2+3}+\frac{1}{1+2+3+4}+\cdots+\frac{1}{1+2+3+\cdots+n}\right)$

យើងមានទំនាក់ទំនង $1+2+3+\cdots+k=\frac{k(k+1)}{2}$

សមមូលឱ្យ $\frac{1}{1+2+3+\cdots+k}=\frac{2}{k(k+1)}=2\left(\frac{1}{k}-\frac{1}{k+1}\right)$

បើ $k=2$ នាំឱ្យ $\frac{1}{1+2}=2\left(\frac{1}{2}-\frac{1}{3}\right)$

បើ $k=3$ នាំឱ្យ $\frac{1}{1+2+3}=2\left(\frac{1}{3}-\frac{1}{4}\right)$

បើ $k=4$ នាំឱ្យ $\frac{1}{1+2+3+4}=2\left(\frac{1}{4}-\frac{1}{5}\right)$

....

....

....

បើ $k=n-1$ នាំឱ្យ $\frac{1}{1+2+3+\cdots+(n-1)}=2\left(\frac{1}{n-1}-\frac{1}{n}\right)$

បើ $k=n$ នាំឱ្យ $\frac{1}{1+2+3+\cdots+n}=2\left(\frac{1}{n}-\frac{1}{n+1}\right)$

បូកសមភាពទាំងពីបញ្ចូលគ្នាយើងបាន៖

$$\frac{1}{1+2}+\frac{1}{1+2+3}+\cdots+\frac{1}{1+2+3+\cdots+n}=2\left(\frac{1}{2}-\frac{1}{3}\right)+2\left(\frac{1}{3}-\frac{1}{4}\right)+2\left(\frac{1}{4}-\frac{1}{5}\right)+$$

$$\cdots+2\left(\frac{1}{n-1}-\frac{1}{n}\right)+2\left(\frac{1}{n}-\frac{1}{n+1}\right)$$

$$=2\left(\frac{1}{2}-\frac{1}{n+1}\right)$$

សមមូលឱ្យ $\lim\limits_{n\to+\infty}\left(\frac{1}{1+2}+\frac{1}{1+2+3}+\cdots+\frac{1}{1+2+3+\cdots+n}\right)=2\lim\limits_{n\to+\infty}\left(\frac{1}{2}-\frac{1}{n+1}\right)$

យើងបាន $L=2\times\frac{1}{2}=1$

ដូចនេះ: $L=1$ ។

021. ចូរគណនាលីមីត $L=\lim\limits_{x\to 0}\left[\dfrac{(x+1)(x+2)(x+3)\ldots(x+n)-n!}{(x+1)(x+2)(x+3)\cdots(x+m)-m!}\right]$ $;\forall n,m\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to 0}\left[\dfrac{(x+1)(x+2)(x+3)\ldots(x+n)-n!}{(x+1)(x+2)(x+3)\cdots(x+m)-m!}\right]$ $;\forall n,m\in\mathbb{N}$

$$=\frac{n!}{m!}\lim_{x\to 0}\left[\frac{\dfrac{(x+1)(x+2)(x+3)\ldots(x+n)}{n!}-1}{\dfrac{(x+1)(x+2)(x+3)\cdots(x+m)}{m!}-1}\right]$$

$$=\frac{n!}{m!}\lim_{x\to 0}\left[\frac{e^{\log\left(\frac{(x+1)(x+2)(x+3)\ldots(x+n)}{n!}\right)}-1}{e^{\log\left(\frac{(x+1)(x+2)(x+3)\ldots(x+m)}{m!}\right)}-1}\right]$$

, ប្រើ $\lim\limits_{x\to a}\dfrac{e^{f(x)}-1}{f(x)}=1,\ \ f(a)=0$

យើងបាន $$L=\frac{n!}{m!}\lim_{x\to 0}\left[\frac{\log\left(\dfrac{(x+1)(x+2)(x+3)\ldots(x+n)}{1\cdot 2\cdot 3\cdots n}\right)}{\log\left(\dfrac{(x+1)(x+2)(x+3)\cdots(x+m)}{1\cdot 2\cdot 3\cdots m}\right)}\right]$$

$$=\frac{n!}{m!}\lim_{x\to 0}\left[\frac{\log\left(\dfrac{(1+x)}{1}\times\dfrac{(2+x)}{2}\times\dfrac{(3+x)}{3}\times\ldots\times\dfrac{(n+x)}{n}\right)}{\log\left(\dfrac{(1+x)}{1}\times\dfrac{(2+x)}{2}\times\dfrac{(3+x)}{3}\times\ldots\times\dfrac{(m+x)}{m}\right)}\right]$$

$$=\frac{n!}{m!}\lim_{x\to 0}\left[\frac{\log(1+x)+\log\left(1+\dfrac{x}{2}\right)+\log\left(1+\dfrac{x}{3}\right)+\cdots+\log\left(1+\dfrac{x}{n}\right)}{\log(1+x)+\log\left(1+\dfrac{x}{2}\right)+\log\left(1+\dfrac{x}{3}\right)+\cdots+\log\left(1+\dfrac{x}{m}\right)}\right]$$

$$=\frac{n!}{m!}\lim_{x\to 0}\left(\frac{x+\dfrac{x}{2}+\dfrac{x}{3}+\cdots+\dfrac{x}{n}}{x+\dfrac{x}{2}+\dfrac{x}{3}+\cdots+\dfrac{x}{m}}\right)$$

, ព្រោះ $\lim\limits_{x\to a}\dfrac{\log[1+f(x)]}{f(x)}=1,\ \ f(a)=0$

$$=\frac{n!}{m!}\cdot\frac{1+\dfrac{1}{2}+\dfrac{1}{3}+\cdots+\dfrac{1}{n}}{1+\dfrac{1}{2}+\dfrac{1}{3}+\cdots+\dfrac{1}{m}}=\frac{n!}{m!}\cdot\frac{H_n}{H_m}$$

ដូចនេះ $L=\dfrac{n!H_n}{m!H_m}$ ។

022. ចូរគណនាលីមីត $L=\lim\limits_{x\to0^+}\left[x\left(\left\lfloor\frac{1}{x}\right\rfloor+\left\lfloor\frac{2}{x}\right\rfloor+\left\lfloor\frac{3}{x}\right\rfloor+\cdots+\left\lfloor\frac{k}{x}\right\rfloor\right)\right]$ $,k\in\mathbb{N}$

ដំណោះស្រាយ

ដោយ $y-1\le\lfloor y\rfloor\le y$ ដែល y ជាចំនួនពិត

យើងបាន $\frac{1}{x}+\frac{2}{x}+\frac{3}{x}+\cdots+\frac{k}{x}-k\le\left\lfloor\frac{1}{x}\right\rfloor+\left\lfloor\frac{2}{x}\right\rfloor+\left\lfloor\frac{3}{x}\right\rfloor+\cdots+\left\lfloor\frac{k}{x}\right\rfloor\le\frac{1}{x}+\frac{2}{x}+\frac{3}{x}+\cdots+\frac{k}{x}$

សមមូល $1+2+3+\cdots+k-kx\le x\left(\left\lfloor\frac{1}{x}\right\rfloor+\left\lfloor\frac{2}{x}\right\rfloor+\left\lfloor\frac{3}{x}\right\rfloor+\cdots+\left\lfloor\frac{k}{x}\right\rfloor\right)\le1+2+3+\cdots+k$

$$\lim_{x\to0^+}\left(\frac{k(k+1)}{2}-kx\right)\le\lim_{x\to0^+}\left[x\left(\left\lfloor\frac{1}{x}\right\rfloor+\left\lfloor\frac{2}{x}\right\rfloor+\left\lfloor\frac{3}{x}\right\rfloor+\cdots+\left\lfloor\frac{k}{x}\right\rfloor\right)\right]\le\lim_{x\to0^+}\left(\frac{k(k+1)}{2}\right)$$

យើងបាន $\frac{k(k+1)}{2}\le L\le\frac{k(k+1)}{2}$ នាំឱ្យ $L=\frac{k(k+1)}{2}$

ដូចនេះ: $L=\frac{k(k+1)}{2}$ ។

023. ចូរគណនាលីមីត $L=\lim\limits_{x\to1}\left(\frac{x+x^2+x^3+\cdots+x^n-n}{x-1}\right)$ $,k\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to1}\left(\frac{x+x^2+x^3+\cdots+x^n-n}{x-1}\right)$ $,k\in\mathbb{N}$

$$=\lim_{x\to1}\left(\frac{x-1+x^2-1+x^3-1+\cdots+x^n-1}{x-1}\right)$$

$$=\lim_{x\to1}\left(\frac{x-1}{x-1}+\frac{x^2-1}{x-1}+\frac{x^3-1}{x-1}+\cdots+\frac{x^n-1}{x-1}\right)$$

$$=\lim_{x\to1}\left(1+\frac{(x-1)(x+1)}{x-1}+\cdots+\frac{(x-1)(x^{n-1}+x^{n-2}+\cdots+x+1)}{x-1}\right)$$

$$=\lim_{x\to1}\left(1+(x+1)+(x^2+x+1)+\cdots+(x^{n-1}+x^{n-2}+\cdots+x+1)\right)$$

$$=(1+2+3+\cdots+n)=\frac{n(n+1)}{2}$$

ដូចនេះ: $L=\frac{n(n+1)}{2}$ ។

024. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1\times3}+\frac{1}{3\times5}+\frac{1}{5\times7}+\cdots+\frac{1}{(2n-1)(2n+1)}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1\times3}+\frac{1}{3\times5}+\frac{1}{5\times7}+\cdots+\frac{1}{(2n-1)(2n+1)}\right)$

$=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\frac{1}{(2k-1)(2k+1)}\right)\right]$, តែ $\frac{1}{(2k-1)(2k+1)}=\frac{1}{2}\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)$

យើងបាន $L=\frac{1}{2}\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\frac{1}{2n-1}-\frac{1}{2n+1}\right)\right]$

$=\frac{1}{2}\lim\limits_{n\to+\infty}\left(1-\frac{1}{2n+1}\right)=\frac{1}{2}$

ដូចនេះ: $L=\frac{1}{2}$ ។

025. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1\cdot4\cdot7}+\frac{1}{4\cdot7\cdot10}+\cdots+\frac{1}{(3n-2)(3n+1)(3n+4)}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1\cdot4\cdot7}+\frac{1}{4\cdot7\cdot10}+\cdots+\frac{1}{(3n-2)(3n+1)(3n+4)}\right)$

$=\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\frac{1}{(3k-2)(3k+1)(3k+4)}\right)\right]$

ដោយ $\frac{1}{(3k-2)(3k+1)(3k+4)}=\frac{1}{6}\cdot\frac{(3k+4)-(3k-2)}{(3k-2)(3k+1)(3k+4)}$

$=\frac{1}{6}\cdot\left(\frac{1}{(3k-2)(3k+1)}-\frac{1}{(3k+1)(3k+4)}\right)$

នោះយើងបាន $L=\frac{1}{6}\lim\limits_{n\to+\infty}\left[\sum\limits_{k=1}^{n}\left(\frac{1}{(3k-2)(3k+1)}-\frac{1}{(3k+1)(3k+4)}\right)\right]$

$=\frac{1}{6}\lim\limits_{n\to+\infty}\left[\frac{1}{1.4}-\frac{1}{(3n+1)(n+4)}\right]$

$=\frac{1}{6}\left(\frac{1}{4}-1\right)=\frac{1}{24}$

ដូចនេះ: $L=\frac{1}{24}$ ។

026. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\cdots\left(1-\frac{1}{n^2}\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\left(1-\frac{1}{2^2}\right)\left(1-\frac{1}{3^2}\right)\left(1-\frac{1}{4^2}\right)\cdots\left(1-\frac{1}{n^2}\right)\right]$

$$=\lim_{n\to+\infty}\left[\left(\frac{2^2-1}{2^2}\right)\left(\frac{3^2-1}{3^2}\right)\left(\frac{4^2-1}{4^2}\right)\cdots\left(\frac{n^2-1}{n^2}\right)\right]$$

$$=\lim_{n\to+\infty}\left[\left(\frac{1\cdot 3}{2^2}\right)\left(\frac{2\cdot 4}{3^2}\right)\left(\frac{3\cdot 5}{4^2}\right)\cdots\left(\frac{(n-1)(n+1)}{n^2}\right)\right]$$

$$=\lim_{n\to+\infty}\left(\frac{1\cdot 2\cdot 3^2\cdot 4^2\cdots(n^2-1)}{2^2\cdot 3^2\cdot 4^2\cdots n^2}\right)$$

$$=\lim_{n\to+\infty}\left(\frac{n^2-1}{2n^2}\right)=\frac{1}{2}$$

ដូចនេះ: $L=\frac{1}{2}$ ។

027. ចូរគណនាលីមីត $L=\lim\limits_{x\to 0}\left(\frac{\sqrt{(1-\cos(x))\sqrt{(1-\cos(x))\sqrt{(1-\cos(x))\cdots}}}}{x^2}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to 0}\left(\frac{\sqrt{(1-\cos(x))\sqrt{(1-\cos(x))\sqrt{(1-\cos(x))\cdots}}}}{x^2}\right)$

$$=\lim_{x\to 0}\left(\frac{(1-\cos(x))^{\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots}}{x^2}\right)$$

$$\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots=\frac{1}{2}\cdot\frac{1}{1-\frac{1}{2}}=1$$

$$=\lim_{x\to 0}\left(\frac{1-\cos(x)}{x^2}\right)$$

$$=\frac{1}{2}$$

ដូចនេះ: $L=\frac{1}{2}$ ។

028. ចូរគណនាលីមីត $L=\lim\limits_{x\to+\infty}\left(\dfrac{x^{100}+(x+1)^{100}+(x+2)^{100}+\cdots+(x+100)^{100}}{x^{100}+100^{100}}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to+\infty}\left(\dfrac{x^{100}+(x+1)^{100}+(x+2)^{100}+\cdots+(x+100)^{100}}{x^{100}+100^{100}}\right)$

$$=\lim_{x\to+\infty}\left[\frac{x^{100}+\left[x\left(1+\frac{1}{x}\right)\right]^{100}+\left[x\left(1+\frac{2}{x}\right)\right]^{100}+\cdots+\left[x\left(1+\frac{100}{x}\right)\right]^{100}}{x^{100}+100^{100}}\right]$$

$$=\lim_{x\to+\infty}\left[\frac{x^{100}\left(1+\left(1+\frac{1}{x}\right)^{100}+\left(1+\frac{2}{x}\right)^{100}+\cdots+\left(1+\frac{100}{x}\right)^{100}\right)}{x^{100}\left(1+\frac{100^{100}}{x^{100}}\right)}\right]$$

$$=\frac{\left(1+(1+0)^{100}+(1+0)^{100}+\cdots+(1+0)^{100}\right)}{(1+0)}=\overbrace{1+1+1+\cdots+1}^{100}=100$$

ដូចនេះ $L=100$ ។

029. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\dfrac{1}{\sqrt{n}}\left(\dfrac{1}{\sqrt{1}+\sqrt{2}}+\dfrac{1}{\sqrt{2}+\sqrt{3}}+\cdots+\dfrac{1}{\sqrt{n}+\sqrt{n+1}}\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\dfrac{1}{\sqrt{n}}\left(\dfrac{1}{\sqrt{1}+\sqrt{2}}+\dfrac{1}{\sqrt{2}+\sqrt{3}}+\cdots+\dfrac{1}{\sqrt{n}+\sqrt{n+1}}\right)\right]$

$$=\lim_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\sum_{k=1}^{n}\left(\frac{1}{\sqrt{k}+\sqrt{k+1}}\right)\right]$$

$$=\lim_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\sum_{k=1}^{n}\left(\sqrt{k+1}-\sqrt{k}\right)\right]$$

$$=\lim_{n\to+\infty}\left[\frac{1}{\sqrt{n}}\left(\sqrt{n+1}-1\right)\right]$$

$$=\lim_{n\to+\infty}\left(\sqrt{1+\frac{1}{n}}-\sqrt{\frac{1}{n}}\right)=1$$

ដូចនេះ $L=1$ ។

030. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\left(2\ln(2)-\ln(3)\right)+\left(2\ln(3)-\ln(8)\right)+\cdots+\left(2\ln(n+1)-\ln(n^2+2n)\right)\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left[\left(2\ln(2)-\ln(3)\right)+\left(2\ln(3)-\ln(8)\right)+\cdots+\left(2\ln(n+1)-\ln(n^2+2n)\right)\right]$

យើងមាន៖ $2\ln(n+1)-\ln(n^2+2n)=2\ln(n+1)-\ln(n)-\ln(n+2)$

$$=\left(\ln(n+1)-\ln(n)\right)+\left(\ln(n+1)-\ln(n+2)\right)$$

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left\{\sum_{k=1}^{n}\left[\left(\ln(n+1)-\ln(n)\right)+\left(\ln(n+1)-\ln(n+2)\right)\right]\right\}$

$$=\lim_{n\to+\infty}\left[\ln(n+1)-\ln(1)+\ln(2)-\ln(n+2)\right]$$

$$=\lim_{n\to+\infty}\left[\ln\left(\frac{n+1}{n+2}\right)+\ln(2)\right]$$

$$=\ln(2)+\ln(1)=\ln(2)$$

ដូចនេះ៖ $L=\ln(2)$ ។

031. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{n}{n^2+1}+\frac{n}{n^2+2}+\frac{n}{n^2+3}+\cdots+\frac{n}{n^2+n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{n}{n^2+1}+\frac{n}{n^2+2}+\frac{n}{n^2+3}+\cdots+\frac{n}{n^2+n}\right)$

$$=\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{n}{n^2+k}\right)\right]$$

ចំពោះគ្រប់ $k\in\mathbb{N}\,,k\le n$ ដែល $n\in\mathbb{N}$

យើងបាន $n^2+1\le n^2+k\le n^2+n$

សមមូលឱ្យ $\frac{n}{n^2+n}\le\frac{n}{n^2+k}\le\frac{n}{n^2+1}$

$$\sum_{k=1}^{n}\left(\frac{n}{n^2+n}\right)\le\sum_{k=1}^{n}\left(\frac{n}{n^2+k}\right)\le\sum_{k=1}^{n}\left(\frac{n}{n^2+1}\right)$$

$$\lim_{n\to+\infty}\left(\frac{n^2}{n^2+n}\right)\le\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{n}{n^2+k}\right)\right]\le\lim_{n\to+\infty}\left(\frac{n^2}{n^2+1}\right)$$

យើងបាន $1\le L\le 1$

យើងទាញបាន $L=1$

ដូចនេះ៖ $L=1$ ។

032. ចូរគណនាលីមីត $L=\lim\limits_{x\to+\infty}\left(\sqrt{x^2+2x}+\sqrt{x^2+4x}+\cdots+\sqrt{x^2+2nx}-nx\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to+\infty}\left(\sqrt{x^2+2x}+\sqrt{x^2+4x}+\cdots+\sqrt{x^2+2nx}-nx\right)$

$$=\lim_{x\to+\infty}\left(\sqrt{x^2+2x}-x+\sqrt{x^2+4x}-x+\cdots+\sqrt{x^2+2nx}-x\right)$$

$$=\lim_{x\to+\infty}\left(\frac{x^2+2x-x^2}{\sqrt{x^2+2x}+x}+\frac{x^2+4x-x^2}{\sqrt{x^2+4x}+x}+\cdots+\frac{x^2+2nx-x^2}{\sqrt{x^2+2nx}+x}\right)$$

$$=\lim_{x\to+\infty}\left(\frac{2x}{|x|\sqrt{1+{}^{2}\!/_{x}}+x}+\frac{4x}{|x|\sqrt{1+{}^{4}\!/_{x}}+x}+\cdots+\frac{2nx}{|x|\sqrt{1+{}^{2n}\!/_{x}}+x}\right)$$

ដោយ $x\to+\infty$ នោះ $|x|=x$

នោះយើងបាន $L=\lim\limits_{x\to+\infty}\left(\frac{2x}{x\left(\sqrt{1+{}^{2}\!/_{x}}+1\right)}+\frac{4x}{x\left(\sqrt{1+{}^{2}\!/_{x}}+1\right)}+\cdots+\frac{2nx}{x\left(\sqrt{1+{}^{2}\!/_{x}}+1\right)}\right)$

$$=\left(\frac{2}{\left(\sqrt{1+0}+1\right)}+\frac{4}{\left(\sqrt{1+0}+1\right)}+\cdots+\frac{2n}{\left(\sqrt{1+0}+1\right)}\right)=\frac{n(n+1)}{2}$$

ដូចនេះ $L=\dfrac{n(n+1)}{2}$ ។

033. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{n^2}\cdot\frac{1+2+3+\cdots+n}{1+\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{n^2}\cdot\frac{1+2+3+\cdots+n}{1+\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}}\right)$

យើងមាន៖ $1+2+3+\cdots+n=\dfrac{n(n+1)}{2}$

ហើយ $1+\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}=\dfrac{\left[1-\left({}^{1}\!/_{2}\right)^{n+1}\right]}{1-{}^{1}\!/_{2}}=2-\dfrac{1}{2^n}$ (តាមរូបមន្តស៊ីតធរណីមាត្រ)

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left(\dfrac{1}{n^2}\cdot\dfrac{\frac{n(n+1)}{2}}{2-\frac{1}{2^n}}\right)=\dfrac{1}{2}\lim\limits_{n\to+\infty}\left(\dfrac{1}{n^2}\cdot\dfrac{n(n+1)}{2-\frac{1}{2^n}}\right)=\dfrac{1}{2}\cdot\dfrac{1}{2-0}=\dfrac{1}{4}$

ដូចនេះ៖ $L=\dfrac{1}{4}$ ។

034. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1+\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}}{1+\frac{1}{3}+\frac{1}{9}+\cdots+\frac{1}{3^n}}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1+\frac{1}{2}+\frac{1}{4}+\cdots+\frac{1}{2^n}}{1+\frac{1}{3}+\frac{1}{9}+\cdots+\frac{1}{3^n}}\right)$

ដោយ $1+\dfrac{1}{2}+\dfrac{1}{4}+\cdots+\dfrac{1}{2^n}=2-\dfrac{1}{2^n}$ និង $1+\dfrac{1}{3}+\dfrac{1}{9}+\cdots+\dfrac{1}{3^n}=\dfrac{1}{2}\left(3-\dfrac{1}{3^n}\right)$

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left(\dfrac{2-\frac{1}{2^n}}{\frac{1}{2}\left(3-\frac{1}{3^n}\right)}\right)=\dfrac{2-0}{\frac{1}{2}(3-0)}=\dfrac{4}{3}$

ដូចនេះ៖ $L=\dfrac{4}{3}$ ។

035. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1\cdot2+2\cdot3+3\cdot4+\cdots+(n-1)n}{n^3}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1\cdot2+2\cdot3+3\cdot4+\cdots+(n-1)n}{n^3}\right)$

ដោយ $1\cdot2+2\cdot3+3\cdot4+\cdots+(n-1)n=\sum\limits_{k=2}^{n}\left[(k-1)k\right]=\sum\limits_{k=2}^{n}(k^2)-\sum\limits_{k=2}^{n}(k)$

$$=\frac{n(n+1)(2n+1)}{6}-\frac{n(n+1)}{2}=\frac{n(n+1)(n-1)}{3}$$

នោះយើងបាន៖ $L=\dfrac{1}{3}\lim\limits_{n\to+\infty}\left(\dfrac{n(n+1)(n-1)}{n^3}\right)=\dfrac{1}{3}\times1=\dfrac{1}{3}$

ដូចនេះ៖ $L=\dfrac{1}{3}$ ។

036. ចូរគណនាលីមីត $L=\lim\limits_{n\to\infty}\left(\dfrac{1-2+3-4+\cdot\cdot+(2n-1)-2n}{\sqrt{n^2+1}}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to\infty}\left(\dfrac{1-2+3-4+\cdot\cdot+(2n-1)-2n}{\sqrt{n^2+1}}\right)$

ដោយ $1-2+3-4+\cdot\cdot+(2n-1)-2n=1+2+3+4+\cdots+2n-4(1+2+3+\cdots+n)$

$$=\frac{2n(2n+1)}{2}-4\cdot\frac{n(n+1)}{2}=-n$$

យើងបាន៖ $L=\lim\limits_{n\to\infty}\left(\dfrac{-n}{\sqrt{n^2+1}}\right)=\lim\limits_{n\to\infty}\left(\dfrac{-n}{|n|\sqrt{1+{}^1\!/_n}}\right)$

+ បើ $n\to+\infty$ នោះ $|n|=n$

+ បើ $n\to-\infty$ នោះ $|n|=-n$

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left(\dfrac{-n}{n\sqrt{1+{}^1\!/_n}}\right)=-1$

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left(\dfrac{-n}{-n\sqrt{1+{}^1\!/_n}}\right)=1$

ដូចនេះ៖ $L=\begin{cases}1 & ,n\to-\infty\\ -1 & ,n\to+\infty\end{cases}$ ។

037. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\sqrt[n]{x_1^n+x_2^n+x_3^n+\cdots+x_{2024}^n}\right)$ ដែល $x_k=\sum\limits_{j=1}^{k}\left(\dfrac{j}{(j+1)!}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\sqrt[n]{x_1^n+x_2^n+x_3^n+\cdots+x_{2024}^n}\right)$

ដោយ $x_k=\sum\limits_{j=1}^{k}\left(\dfrac{j}{(j+1)!}\right)$ នាំឱ្យ $x_{k+1}=\sum\limits_{j=1}^{k}\left(\dfrac{j+1}{(j+2)!}\right)$

នោះ៖ $x_{k+1}-x_k=\dfrac{k+1}{(k+2)!}>0$ គេបាន $x_{k+1}>x_k\quad,\forall x\in\mathbb{N}$

យើងទាញបាន៖ $x_{2024}^n<x_1^n+x_2^n+x_3^n+\cdots+x_{2024}^n<2024\cdot x_{2024}^n$

សមមូលឱ្យ $x_{2024}<\sqrt[n]{x_1^n+x_2^n+x_3^n+\cdots+x_{2024}^n}<\sqrt[n]{2024}\cdot x_{2024}\quad(*)$

ម្យ៉ាង $x_k=\sum_{j=1}^{k}\left(\frac{j}{(j+1)!}\right)=\sum_{j=1}^{k}\left(\frac{1}{(j)!}-\frac{1}{(j+1)!}\right)=1-\frac{1}{(j+1)!}$ យក $x=2024$

នោះ៖ $x_{2024}=1-\frac{1}{2025!}$

តាម (*) យើងបាន៖ $\left(1-\frac{1}{2025!}\right)<\sqrt[n]{x_1^n+x_2^n+x_3^n\cdots+x_{2024}^n}<2024^{\frac{1}{n}}\left(1-\frac{1}{2025!}\right)$

សមមូល $\lim\limits_{n\to+\infty}\left(1-\frac{1}{2025!}\right)<\lim\limits_{n\to+\infty}\left(\sqrt[n]{x_1^n+x_2^n+x_3^n\cdots+x_{2024}^n}\right)<\lim\limits_{n\to+\infty}\left[2024^{\frac{1}{n}}\left(1-\frac{1}{2025!}\right)\right]$

$$\left(1-\frac{1}{2025!}\right)<L<\left(1-\frac{1}{2025!}\right)\Rightarrow L=\left(1-\frac{1}{2025!}\right)$$

ដូចនេះ៖ $L=1-\frac{1}{2025!}$ ។

038. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1^2+3^2+5^2+\cdots+(2n-1)^2}{n^3}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1^2+3^2+5^2+\cdots+(2n-1)^2}{n^3}\right)$

ដោយ $1^2+3^2+5^2+\cdots+(2n-1)^2=\sum_{k=1}^{n}(2k-1)^2$

$$=\sum_{k=1}^{n}\left(4k^2-4k+1\right)$$

$$=4\sum_{k=1}^{n}(k^2)-4\sum_{k=1}^{n}(k)+\sum_{k=1}^{n}(1)$$

$$=4\cdot\frac{n(n+1)(2n+1)}{6}-4\cdot\frac{n(n+1)}{2}-n$$

$$=\frac{1}{3}(4n^3-n)$$

នោះយើងបាន៖ $L=\frac{1}{3}\lim\limits_{n\to+\infty}\left(\frac{4n^3-n}{n^3}\right)=\frac{4}{3}$

ដូចនេះ៖ $L=\frac{4}{3}$ ។

038. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1\cdot2+2\cdot3+3\cdot4+\cdots+n(n+1)}{n^3}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1\cdot2+2\cdot3+3\cdot4+\cdots+n(n+1)}{n^3}\right)$

ដោយ $1\cdot2+2\cdot3+3\cdot4+\cdots+n(n+1)=\sum\limits_{k=2}^{n}\left[k(k+1)\right]=\sum\limits_{k=2}^{n}(k^2)+\sum\limits_{k=2}^{n}(k)$

$$=\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}$$

$$=\frac{1}{3}(n^3+3n^2+2n)$$

នោះយើងបាន៖ $L=\dfrac{1}{3}\lim\limits_{n\to+\infty}\left(\dfrac{n^3+3n^2+2n}{n^3}\right)=\dfrac{1}{3}$

ដូចនេះ៖ $L=\dfrac{1}{3}$ ។

038. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1+11+111+1111+\cdots+\overbrace{111\cdots111}^{n}}{10^n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{1+11+111+1111+\cdots+\overbrace{111\cdots111}^{n}}{10^n}\right)$

ដោយ $1+11+111+1111+\cdots+\overbrace{111\cdots111}^{n}=\dfrac{1}{9}\left(9+99+999+9999+\cdots+\overbrace{999\cdots999}^{n}\right)$

$$=\frac{1}{9}\left(10-1+10^2-1+10^3-1+\cdots+10^n-1\right)$$

$$=\frac{1}{9}\left(10+10^2+10^3+\cdots+10^n\overbrace{-1-1-\cdots-1}^{n}\right)$$

$$=\frac{1}{9}\left(10\cdot\frac{10^n-1}{10-1}-n\right)$$

$$=\frac{1}{81}\cdot\left(10\cdot10^n-9n-10\right)$$

នោះយើងបាន៖ $L=\frac{1}{81}\lim_{n\to+\infty}\left(\frac{10\cdot10^n-9n-10}{10^n}\right)$

$$=\frac{1}{81}\lim_{n\to+\infty}\left(10-\frac{9n}{10^n}-\frac{10}{10^n}\right)=\frac{1}{81}(10-0-0)=\frac{10}{81}$$

ដូចនេះ៖ $L=\frac{10}{81}$ ៕

039. ចូរគណនាលីមីត $L=\lim_{x\to1}\left(\frac{x^n-nx+n-1}{(x-1)^2}\right)\ ,\forall n\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim_{x\to1}\left(\frac{x^n-nx+n-1}{(x-1)^2}\right)\ ,\forall n\in\mathbb{N}$

$$=\lim_{x\to1}\left(\frac{x^n-1-nx+n}{(x-1)^2}\right)$$

$$=\lim_{x\to1}\left(\frac{(x-1)(1+x+x^2+\cdots+x^{n-2}+x^{n-1})-n(x-1)}{(x-1)^2}\right)$$

$$=\lim_{x\to1}\left(\frac{1+x+x^2+\cdots+x^{n-2}+x^{n-1}-n}{(x-1)}\right)$$

$$=\lim_{x\to1}\left(\frac{(x-1)+(x^2-1)+\cdots+(x^{n-2}-1)+(x^{n-1}-1)}{(x-1)}\right)$$

$$=\lim_{x\to1}\left(\frac{(x-1)}{(x-1)}+\frac{(x^2-1)}{(x-1)}+\cdots+\frac{(x^{n-2}-1)}{(x-1)}+\frac{(x^{n-1}-1)}{(x-1)}\right)$$

$$=\lim_{x\to1}\left(1+(x+1)+\cdots+(x^{n-3}+x^{n-2}+\cdots+x+1)+(x^{n-2}+x^{n-1}+\cdots+x+1)\right)$$

$$=1+2+3+(n-2)+(n-1)$$

$$=\frac{n(n-1)}{2}$$

ដូចនេះ៖ $L=\frac{n(n-1)}{2}$ ៕

040. ចូរគណនាលីមីត $L=\lim\limits_{x\to 0}\left(\dfrac{1-\cos(x)\cos(2x)\cos(3x)\cdots\cos(nx)}{x^2}\right)$,$\forall n\in\mathbb{N}$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to 0}\left(\dfrac{1-\cos(x)\cos(2x)\cos(3x)\cdots\cos(nx)}{x^2}\right)$,$\forall n\in\mathbb{N}$

$$=\lim_{x\to 0}\left(\frac{1-\cos(x)+\cos(x)-\cos(x)\cos(2x)\cos(3x)\cdots\cos(nx)}{x^2}\right)$$

$$=\lim_{x\to 0}\left(\frac{1-\cos(x)}{x^2}\right)+\lim_{x\to 0}\left(\frac{\cos(x)(1-\cos(2x)\cos(3x)\cdots\cos(nx)}{x^2}\right)$$

$$=\frac{1}{2}+\lim_{x\to 0}\left(\frac{1-\cos(2x)+\cos(2x)-\cos(2x)\cos(3x)\cdots\cos(nx)}{x^2}\right)$$

$$=\frac{1}{2}+\lim_{x\to 0}\left(\frac{1-\cos(2x)}{x^2}\right)+\lim_{x\to 0}\left(\frac{\cos(2x)(1-\cos(3x)\cdots\cos(nx)}{x^2}\right)$$

$$=\frac{1}{2}+\frac{2^2}{2}+\lim_{x\to 0}\left(\frac{1-\cos(3x)+\cos(3x)-\cos(3x)\cos(4x)\cdots\cos(nx)}{x^2}\right)$$

$$=\frac{1}{2}+\frac{2^2}{2}+\lim_{x\to 0}\left(\frac{1-\cos(3x)}{x^2}\right)+\lim_{x\to 0}\left(\frac{\cos(3x)(1-\cos(4x)\cdots\cos(nx)}{x^2}\right)$$

$$=\frac{1}{2}+\frac{2^2}{2}+\frac{3^2}{2}+\lim_{x\to 0}\left(\frac{1-\cos(4x)\cdots\cos(nx)}{x^2}\right)$$

$$=\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$=\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$=\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots$$

$$=\frac{1}{2}+\frac{2^2}{2}+\frac{3^2}{2}+\cdots+\frac{n^2}{2}$$

$$=\frac{1+2^2+3^2+\cdots+n^2}{2}$$

$$=\frac{n(n+1)(2n+1)}{12}$$

ដូចនេះ៖ $L=\dfrac{n(n-1)(2n+1)}{12}$ ។

សម្គាល់៖ $\lim\limits_{x\to 0}\dfrac{1-\cos(nx)}{x^2}=\dfrac{n^2}{2}$

041. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{14+144+1444+14444+\cdots+1\overbrace{444\cdots444}^{n}}{10^n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\dfrac{14+144+1444+14444+\cdots+1\overbrace{444\cdots444}^{n}}{10^n}\right)$

ពិនិត្យ $14+144+1444+14444+\cdots+1\overbrace{444\cdots444}^{n}$

តាង $S=14+144+1444+14444+\cdots+1\overbrace{444\cdots444}^{n}$

$$=10+100+1000+\cdots+1\overbrace{000\cdots000}^{n}+4+44+444+\cdots+\overbrace{444\cdots444}^{n}$$

$$=10+100+1000+\cdots+1\overbrace{000\cdots000}^{n}+\frac{4}{9}\left(10-1+10^2-1++\cdots+10^n-1\right)$$

$$=10+10^2+10^3+\cdots+10^n+\frac{4}{9}\left(10+10+10^2+10^3+\cdots+10^n-n\right)$$

$$=10\cdot\frac{10^n-1}{10-1}+\frac{4}{9}\left(10\cdot\frac{10^n-1}{10-1}-n\right)$$

$$=10\cdot\frac{10^n-1}{10-1}+\frac{4}{9}\times10\cdot\frac{10^n-1}{10-1}-\frac{4n}{9}$$

$$=\left(\frac{130}{81}\right)(10^n-1)-\frac{4n}{9}$$

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left(\dfrac{=\left(\frac{130}{81}\right)(10^n-1)-\frac{4n}{9}}{10^n}\right)$

$$=\lim_{n\to+\infty}\left(\frac{130}{81}\cdot\left(1-\frac{1}{10^n}\right)-\frac{4n}{9\cdot10^n}\right)$$

$$=\frac{130}{81}(1-0)-0=\frac{130}{81}$$

ដូចនេះ៖ $L=\dfrac{130}{81}$ ។

042. ចូរគណនាលីមីត $L=\lim\limits_{n\to\infty}\left[\sum\limits_{k=1}^{n}\left(\dfrac{2k+1}{\sqrt{(k+1)^2+1}-\sqrt{k^2+1}}\right)-\sqrt{n^2+1}\right]$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to\infty}\left[\sum\limits_{k=1}^{n}\left(\dfrac{2k+1}{\sqrt{(k+1)^2+1}-\sqrt{k^2+1}}\right)-\sqrt{n^2+1}\right]$

ពិនិត្យ $\dfrac{2k+1}{\sqrt{(k+1)^2+1}+\sqrt{k^2+1}}=\dfrac{(2k+1)\left(\sqrt{(k+1)^2+1}-\sqrt{k^2+1}\right)}{(k+1)^2+1-k^2-1}$

$$=\sqrt{(k+1)^2+1}-\sqrt{k^2+1}$$

នាំឱ្យ $\sum\limits_{k=1}^{n}\left(\sqrt{(k+1)^2+1}-\sqrt{k^2+1}\right)=\left(\sqrt{(n+1)^2+1}-\sqrt{2}\right)$

នោះយើងបាន៖ $L=\lim\limits_{n\to\infty}\left(\sqrt{(n+1)^2+1}-\sqrt{2}-\sqrt{n^2+1}\right)$

$$=\lim_{n\to\infty}\left(\frac{2n+1}{\sqrt{(n+1)^2+1}+\sqrt{n^2+1}}\right)-\sqrt{2}$$

$$=\lim_{n\to\infty}\left(\frac{2n+1}{|n|\left(\sqrt{(1+\frac{1}{n})^2+\frac{1}{n}}+\sqrt{1+\frac{1}{n^2}}\right)}\right)-\sqrt{2}$$

$$=\lim_{n\to\infty}\left(\frac{2n}{|n|\left(\sqrt{(1+0)^2+0}+\sqrt{1+0}\right)}\right)-\sqrt{2}$$

$$=\lim_{n\to\infty}\left(\frac{n}{|n|}\right)-\sqrt{2}$$

+ បើ $n\to+\infty$ នោះ៖ $|n|=n$ នាំឱ្យ $L=\lim\limits_{n\to+\infty}\left(\dfrac{n}{n}\right)-\sqrt{2}=1-\sqrt{2}$

+ បើ $n\to-\infty$ នោះ៖ $|n|=-n$ នាំឱ្យ $L=\lim\limits_{n\to+\infty}\left(\dfrac{n}{-n}\right)-\sqrt{2}=-1-\sqrt{2}$

ដូចនេះ៖ $L=\begin{cases}1-\sqrt{2} & ,n\to-\infty\\ -1-\sqrt{2} & ,n\to+\infty\end{cases}$ ។

043. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^2}+\cdots+\frac{2n-1}{2^n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{2}+\frac{3}{2^2}+\frac{5}{2^2}+\cdots+\frac{2n-1}{2^n}\right)\qquad(!)$

$$\frac{1}{2}L=\lim_{n\to+\infty}\left(\frac{1}{2^2}+\frac{3}{2^3}+\frac{5}{2^4}+\cdots+\frac{2n-1}{2^{n+1}}\right)\qquad(!!)$$

យក $(!)-(!!)$ យើងបាន៖ $\frac{1}{2}L=\lim\limits_{n\to+\infty}\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots+\frac{1}{2^n}-\frac{2n-1}{2^{n+1}}\right)$

$$\Rightarrow L=2\lim_{n\to+\infty}\left(\frac{1}{2}+\frac{1}{2}\cdot\frac{1-\frac{1}{2^n}}{1-\frac{1}{2}}-\frac{2n-1}{2^{n+1}}\right)$$

$$=2\lim_{n\to+\infty}\left(\frac{1}{2}+1-\frac{1}{2^n}-\frac{2n-1}{2^{n+1}}\right)=2\left(\frac{1}{2}+1-0-0\right)=3$$

ដូចនេះ៖ $L=3$ ។

044. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1\cdot1!+2\cdot2!+3\cdot3!+4\cdot4!+\cdots+n\cdot n!}{(n+1)!}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1\cdot1!+2\cdot2!+3\cdot3!+4\cdot4!+\cdots+n\cdot n!}{(n+1)!}\right)$

តាង $S=1\cdot1!+2\cdot2!+3\cdot3!+4\cdot4!+\cdots+n\cdot n!=\sum\limits_{k=1}^{n}(k\cdot k!)$

ប៉ុន្តែ $k\cdot k!=(k+1-1)k!=(k+1)!-k!$

នោះ៖ $S=\sum\limits_{k=1}^{n}(k\cdot k!)=\sum\limits_{k=1}^{n}[(k+1)!-k!]$

$$=(n+1)!-1!=(n+1)!-1$$

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left(\frac{(n+1)!-1}{(n+1)!}\right)=1$

ដូចនេះ៖ $L=1$ ។

045. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{12}{(4-3)(4^2-3^2)}+\frac{12^2}{(4^2-3^2)(4^3-3^3)}+\cdots+\frac{12^n}{(4^n-3^n)(4^{n+1}-3^{n+1})}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{12}{(4-3)(4^2-3^2)}+\frac{12^2}{(4^2-3^2)(4^3-3^3)}+\cdots+\frac{12^n}{(4^n-3^n)(4^{n+1}-3^{n+1})}\right)$

តាង $S=\frac{12}{(4-3)(4^2-3^2)}+\frac{12^2}{(4^2-3^2)(4^3-3^3)}+\cdots+\frac{12^n}{(4^n-3^n)(4^{n+1}-3^{n+1})}$

យើងបាន $S=\sum\limits_{k=1}^{n}\left(\frac{12^k}{(4^k-3^k)(4^{k+1}-3^{k+1})}\right)$

ពិនិត្យ $$\frac{12^k}{(4^k-3^k)(4^{k+1}-3^{k+1})}=\frac{4^k(4^{k+1}-3^{k+1})-4^{k+1}(4^k-3^k)}{(4^k-3^k)(4^{k+1}-3^{k+1})}$$
$$=\frac{4^k}{(4^k-3^k)(4^{k+1}-3^{k+1})}-\frac{4^{k+1}}{(4^{k+1}-3^{k+1})}$$

យើងបាន $S=\sum\limits_{k=1}^{n}\left(\frac{4^k}{(4^k-3^k)(4^{k+1}-3^{k+1})}-\frac{4^{k+1}}{(4^{k+1}-3^{k+1})}\right)=4-\frac{4^{n+1}}{4^{n+1}-3^{n+1}}$

នោះយើងបាន៖ $L=\lim\limits_{n\to+\infty}\left(4-\frac{4^{n+1}}{4^{n+1}-3^{n+1}}\right)=\lim\limits_{n\to+\infty}\left(4-\frac{4^{n+1}}{4^{n+1}}\right)=3$

ដូចនេះ៖ $L=3$ ។

046. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\sqrt{2n-1}-\frac{1}{\sqrt{3+2\sqrt{2}}}-\frac{1}{\sqrt{5+2\sqrt{6}}}-\cdots-\frac{1}{\sqrt{2n-1+2\sqrt{n^2-n}}}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\sqrt{2n-1}-\frac{1}{\sqrt{3+2\sqrt{2}}}-\frac{1}{\sqrt{5+2\sqrt{6}}}-\cdots-\frac{1}{\sqrt{2n-1+2\sqrt{n^2-n}}}\right)$
$$=\lim_{n\to+\infty}\left[\sqrt{2n-1}-\sum_{k=2}^{n}\left(\frac{1}{\sqrt{2k-1+2\sqrt{k^2-k}}}\right)\right]$$

តាង $S=\sum\limits_{k=1}^{n}\left(\frac{1}{\sqrt{2k-1+2\sqrt{k^2-k}}}\right)$

ប៉ុន្តែ $\dfrac{1}{\sqrt{2k-1+2\sqrt{k^2-k}}}=\dfrac{1}{\sqrt{\sqrt{k^2}+2\sqrt{k(k-1)}+\sqrt{(k-1)^2}}}$

$$=\frac{1}{\sqrt{k}+\sqrt{k-1}}=\sqrt{k}-\sqrt{k-1}$$

យើងបាន $S=\sum_{k=2}^{n}\left(\sqrt{k}-\sqrt{k-1}\right)=\sqrt{n}-1$

នោះយើងបាន៖ $L=\lim_{n\to+\infty}\left[\sqrt{2n-1}-(\sqrt{n}-1)\right]=1+\lim_{n\to+\infty}\left(\sqrt{2n-1}-\sqrt{n}\right)=\dfrac{2+\sqrt{2}}{1+\sqrt{2}}$

ដូចនេះ៖ $L=\dfrac{2+\sqrt{2}}{1+\sqrt{2}}$ ។

047. ចូរគណនាលីមីត $L=\lim_{n\to+\infty}\left(\dfrac{1}{2}-\dfrac{2}{2^2}+\dfrac{3}{2^3}-\dfrac{4}{2^4}+\cdots+\dfrac{n}{2^n}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim_{n\to+\infty}\left(\dfrac{1}{2}-\dfrac{2}{2^2}+\dfrac{3}{2^3}-\dfrac{4}{2^4}+\cdots+\dfrac{n}{2^n}\right)$

តាង $S=\dfrac{1}{2}-\dfrac{2}{2^2}+\dfrac{3}{2^3}-\dfrac{4}{2^4}+\cdots+\dfrac{n}{2^n}$ (!)

នោះ៖ $2S=1-\dfrac{2}{2}+\dfrac{3}{2^2}-\dfrac{4}{2^3}+\cdots+\dfrac{n}{2^{n-1}}$ (!!)

យក $(!)+(!!)$ យើងបាន៖ $3S=1-\dfrac{1}{2}+\dfrac{1}{2^2}-\dfrac{1}{2^3}+\cdots-\dfrac{1}{2^{n-2}}+\dfrac{1}{2^{n-1}}+\dfrac{n}{2^n}$

$$\Rightarrow S=\frac{1}{3}\cdot\left[\frac{1-\left(-\frac{1}{2}\right)^n}{1+\frac{1}{2}}+\frac{n}{2^n}\right]=\frac{2}{9}\left[1-\left(-\frac{1}{2}\right)^n\right]+\frac{n}{3\cdot 2^n}$$

នោះយើងបាន៖ $L=\lim_{n\to+\infty}\left[\dfrac{2}{9}\left[1-\left(-\dfrac{1}{2}\right)^n\right]+\dfrac{n}{3\cdot 2^n}\right]$

$$=\frac{2}{9}\lim_{n\to+\infty}\left[1-\left(-\frac{1}{2}\right)^n\right]+\frac{1}{3}\lim_{n\to+\infty}\left(\frac{n}{2^n}\right)=\frac{2}{9}(1-0)+\frac{1}{3}\times 0=\frac{2}{9}$$

ដូចនេះ៖ $L=\dfrac{2}{9}$ ។

048. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{n^2-1}+\frac{2}{n^2-1}+\frac{3}{n^2-1}+\cdots+\frac{n}{n^2-1}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{n^2-1}+\frac{2}{n^2-1}+\frac{3}{n^2-1}+\cdots+\frac{n}{n^2-1}\right)$

$$=\lim_{n\to+\infty}\left(\frac{1+2+3+\cdots+n}{n^2-1}\right)$$

$$=\frac{1}{2}\lim_{n\to+\infty}\left(\frac{n(n+1)}{n^2-1}\right)=\frac{1}{2}$$

$$1+2+3+\cdots+n=\frac{n(n+1)}{2}$$

ដូចនេះ: $L=\frac{1}{2}$ ។

049. ចូរគណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1+\sqrt{n}}+\frac{2}{1+\sqrt{2n}}+\frac{3}{1+\sqrt{2n}}+\cdots+\frac{n}{1+\sqrt{n^2}}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{n\to+\infty}\left(\frac{1}{1+\sqrt{n}}+\frac{2}{1+\sqrt{2n}}+\frac{3}{1+\sqrt{2n}}+\cdots+\frac{n}{1+\sqrt{n^2}}\right)$

$$=\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{k}{1+\sqrt{k^2}}\right)\right]$$

ចំពោះគ្រប់ $k\in\mathbb{N}\,,k\le n$ ដែល $n\in\mathbb{N}$

យើងបាន $\sqrt{n}+1\le\sqrt{kn}+1\le n+1$

សមមូលឱ្យ $\frac{k}{n+1}\le\frac{k}{\sqrt{kn}+1}\le\frac{k}{\sqrt{n}+1}$

$$\sum_{k=1}^{n}\left(\frac{k}{n+1}\right)\le\sum_{k=1}^{n}\left(\frac{k}{\sqrt{kn}+1}\right)\le\sum_{k=1}^{n}\left(\frac{k}{\sqrt{n}+1}\right)$$

$$\frac{1}{2}\lim_{n\to+\infty}\left(\frac{n(n+1)}{n+1}\right)\le\lim_{n\to+\infty}\left[\sum_{k=1}^{n}\left(\frac{k}{\sqrt{kn}+1}\right)\right]\le\frac{1}{2}\lim_{n\to+\infty}\left(\frac{n(n+1)}{\sqrt{n}+1}\right)$$

សមមូលឱ្យ៖ $+\infty\le L\le+\infty$

នោះយើងបាន៖ $L=+\infty$

ដូចនេះ: $L=+\infty$ ។

050. ចូរគណនាលីមីត $L=\lim\limits_{x\to0}\left(\dfrac{1-\sqrt{\cos(x)\cos(2x)\cos(3x)\cdots\cos(n)}}{x^2}\right)$

ដំណោះស្រាយ

គណនាលីមីត $L=\lim\limits_{x\to0}\left(\dfrac{1-\sqrt{\cos(x)\cos(2x)\cos(3x)\cdots\cos(n)}}{x^2}\right)$

តាង $a_n=\lim\limits_{x\to0}\left(\dfrac{1-\sqrt{\cos(x)\cos(2x)\cos(3x)\cdots\cos(n)}}{x^2}\right)$

$$a_n=\lim_{x\to0}\left(\frac{1-\sqrt{\cos(nx)}+\sqrt{\cos(nx)}-\sqrt{\cos(x)\cos(2x)\cos(3x)\cdots\cos(nx)}}{x^2}\right)$$

$$=\lim_{x\to0}\left(\frac{1-\sqrt{\cos(nx)}}{x^2}\right)+\lim_{x\to0}\frac{\sqrt{\cos(nx)}\left(1-\sqrt{\cos(x)\cos(2x)\cos(3x)\cdots\cos(n-1)x}\right)}{x^2}$$

$$=\lim_{x\to0}\left(\frac{1-\cos(nx)}{x^2}\times\frac{1}{1+\sqrt{\cos(nx)}}\right)+\lim_{x\to0}\frac{\left(1-\sqrt{\cos(x)\cos(2x)\cos(3x)\cdots\cos(n-1)x}\right)}{x^2}$$

$$=\frac{n^2}{4}+a_{n-1}$$

នោះ $a_n-a_{n-1}=\dfrac{n^2}{4}$

សមមូលឱ្យ $\sum\limits_{k=2}^{n}(a_k-a_{k-1})=\sum\limits_{k=2}^{n}\left(\dfrac{n^2}{4}\right)$

យើងបាន $a_n-a_1=\dfrac{1}{4}\left(\dfrac{n(n+1)(2n+1)}{6}-1\right)$

តែ $a_1=\lim\limits_{x\to0}\dfrac{1-\sqrt{\cos(x)}}{x^2}$

$$=\lim_{x\to0}\frac{1-\cos(x)}{x^2}\times\frac{1}{1+\sqrt{\cos(x)}}=\frac{1}{4}$$

នោះ $a_n-\dfrac{1}{4}=\dfrac{1}{4}\left(\dfrac{n(n+1)(2n+1)}{6}-1\right)\Rightarrow a_n=\dfrac{n(n+1)(2n+1)}{24}$

ដូចនេះ: $L=\dfrac{n(n+1)(2n+1)}{24}$ ។